

Long-Term Shoreline Evolution of the Chilika Lagoon Coast, India (1975–2026) Using the Digital Shoreline Analysis System (DSAS)

Abstract

Shoreline change is a dynamic coastal process driven by the combined influence of natural hydrodynamic processes and anthropogenic activities. Understanding long-term shoreline evolution is essential for sustainable coastal management, particularly in ecologically sensitive coastal lagoons where shoreline instability threatens biodiversity, infrastructure, and local livelihoods. Chilika Lagoon, the largest brackish-water lagoon in Asia and a Ramsar Wetland of International Importance, has experienced substantial geomorphological changes over recent decades due to wave action, longshore sediment transport, tidal inlet migration, tropical cyclones, and increasing human interventions. However, a comprehensive assessment of long-term shoreline evolution incorporating recent satellite observations and future shoreline projections has remained limited.

This study quantified shoreline dynamics along the Chilika Lagoon coast over a 51-year period (1975–2026) using multi-temporal satellite imagery integrated with the Digital Shoreline Analysis System (DSAS 5.1). Historical shorelines were extracted from Landsat imagery using the Modified Normalized Difference Water Index (MNDWI) implemented within the Google Earth Engine platform and analysed along 591 transects using Net Shoreline Movement (NSM), End Point Rate (EPR), and Linear Regression Rate (LRR).

The results indicate that shoreline evolution is predominantly characterized by accretion, with **479 transects (81.05%)** exhibiting shoreline advancement, whereas **81 transects (13.71%)**

experienced erosion and only **3 transects (0.51%)** remained stable. Zone II showed the highest accretional activity, while localized erosion was concentrated near the tidal inlet and adjacent downdrift sectors, highlighting the strong influence of inlet migration and longshore sediment transport on shoreline behaviour. Mean accretion rates ranged from **0.29 to 1.83 m yr⁻¹**, whereas mean erosion rates varied between **-0.002 and -0.75 m yr⁻¹**. Forecasting results suggest that accretion will continue across most sectors of the barrier coast, although erosion hotspots adjacent to the inlet are likely to persist under future coastal conditions.

The observed shoreline evolution reflects the complex interaction between coastal hydrodynamics, sediment transport, tropical cyclone activity, and increasing anthropogenic modifications, including aquaculture expansion and coastal infrastructure development. This study provides one of the longest continuous assessments of shoreline dynamics for the Chilika Lagoon coast and demonstrates the effectiveness of integrating **Google Earth Engine, MNDWI, and DSAS** for multi-decadal shoreline monitoring and prediction. The findings provide an important scientific basis for coastal vulnerability assessment, climate change adaptation, ecosystem conservation, and sustainable shoreline management along the Odisha coast.

Keyword: Shoreline change; Digital Shoreline Analysis System (DSAS); MNDWI; Landsat; Chilika Lagoon; Coastal Erosion-Accretion.

1 Introduction:

Coastal zones represent one of the most dynamic interfaces between terrestrial and marine environments, where continuous interactions among oceanographic, atmospheric, geological, and biological processes shape coastal landscapes over multiple spatial and temporal scales. Globally, coastlines provide ecosystem services fundamental to ecological sustainability and socio-economic development, supporting fisheries, coastal wetlands, mangrove forests, tourism, maritime transport, agriculture, and industry, while acting as natural buffers against storm surges and wave attack. Approximately 40% of the global population resides within 100 km of the coastline, a proportion expected to increase considerably in coming decades because of urbanization and coastal economic expansion (United Nations, 2023). Maintaining the stability and resilience of coastal environments has therefore become a major scientific and policy priority worldwide.

Coastal systems are among the most vulnerable environments on Earth. Their morphology evolves continuously in response to waves, tides, currents, river discharge, sediment transport, sea-level fluctuations, and extreme weather. Unlike terrestrial landscapes, shorelines rarely remain static, instead exhibiting persistent erosion, accretion, or alternating phases of both depending on local sediment budgets and hydrodynamic conditions (Masselink & Hughes, 2023). Shoreline position therefore represents an integrated response to processes operating over periods ranging from individual storms to millennial-scale geomorphological evolution.

Over recent decades, the rate of shoreline modification has accelerated due to intensifying anthropogenic pressures combined with climate-induced change. Urban expansion, tourism development, port construction, aquaculture, land reclamation, dredging, and shoreline

engineering frequently alter natural sediment transport pathways and reduce coastal ecosystem resilience. Simultaneously, climate change has emerged as a dominant driver of coastal transformation through rising sea levels, warming seas, altered wave climates, and more frequent extreme weather (IPCC, 2023).

Sea-level rise is one of the most significant long-term threats to coastal environments. Global mean sea level has risen at an accelerating rate over the past century, primarily due to thermal expansion of seawater and melting glaciers and ice sheets, and is projected to rise by approximately 0.44–0.76 m by 2100 depending on emission pathway, increasing the frequency of coastal flooding, shoreline retreat, and wetland loss (Fox-Kemper et al., 2021; IPCC, 2023). Low-gradient systems such as barrier islands, estuaries, lagoons, and deltas are especially sensitive, since even small sea-level increases can substantially alter sediment redistribution and shoreline position (Ranasinghe, 2020).

Extreme weather events further amplify coastal vulnerability. Tropical cyclones and storm surges can produce rapid shoreline retreat through intense erosion, dune breaching, and large-scale sediment redistribution; while many beaches recover after individual storms, repeated extreme events often exceed the natural recovery capacity of beaches and barrier systems. Although the global frequency of tropical cyclones may not increase substantially, the proportion of high-intensity storms is projected to rise under continued warming (Knutson et al., 2021; IPCC, 2023). Wave climate is itself shifting in response to climate change, with regional trends in wave height and direction now documented across several ocean basins (Morim et al., 2022), carrying direct implications for longshore sediment transport and shoreline behaviour.

Sediment transport exerts a further fundamental control on shoreline evolution. Along wave-dominated coastlines, sediment is continuously redistributed by longshore currents; where sediment supply exceeds erosional losses, shorelines prograde, while sediment deficits produce recession (Davidson-Arnott et al., 2019; Masselink & Hughes, 2023). Coastal structures such as seawalls, groynes, breakwaters, and jetties frequently interrupt these pathways, producing sediment accumulation updrift and erosion downdrift (Kamphuis, 2020; Ranasinghe, 2020) — a pattern well documented along parts of the Odisha coast (Pradhan et al., 2022).

Land-use change compounds these pressures. Coastal urbanization and the expansion of aquaculture and tourism infrastructure frequently remove mangroves, dunes, and other vegetation that naturally stabilize shorelines and trap sediment, reducing the coast's capacity to withstand wave energy (Friess et al., 2022). River regulation is a further anthropogenic driver: dams and diversions reduce downstream sediment delivery, and several major deltas worldwide have experienced accelerated retreat following upstream sediment reduction (Syvitski et al., 2022).

These global pressures are strongly reflected on the Indian coastline, which extends approximately 7,516 km across highly diverse geomorphological settings. The east coast is especially exposed, facing the Bay of Bengal — one of the most cyclone-prone ocean basins in the world, generating roughly 70% of North Indian Ocean tropical cyclones despite a shallow shelf and densely populated deltaic coast that amplify their impact (India Meteorological Department, 2024).

Odisha, on the northwestern Bay of Bengal, experiences some of the highest cyclone frequencies on the Indian coast; historical records indicate that roughly a third of severe cyclones affecting India make landfall there (India Meteorological Department, 2024). Recent storms — including Phailin (2013), Hudhud (2014), Fani (2019), Yaas (2021), and Dana (2024) — have driven substantial coastal erosion and morphological change along the Odisha coast (Mohanty et al., 2022).

Within this setting, Chilika Lagoon — Asia's largest brackish-water lagoon, designated a Ramsar Wetland of International Importance in 1981 — offers a distinctive natural laboratory for studying long-term shoreline evolution because of its dynamic barrier coast and migrating tidal inlet (Ramsar Convention Secretariat, 2022; Samal, 2011). The lagoon's water-spread area fluctuates seasonally between roughly 906 km² and 1,165 km², fed by more than 52 rivers draining a catchment exceeding 4,300 km² (Samal, 2011). Its approximately 65-km sandy barrier is continuously reshaped by wave action, tidal exchange, and longshore sediment transport, with the inlet migrating in response to sediment availability (Chandramohan & Nayak, 1994). Estimated annual longshore sediment transport along the barrier is on the order of $0.65\text{--}0.89 \times 10^6 \text{ m}^3$ (Chandramohan & Nayak, 1994; Pradhan et al., 2017), underscoring the dominant role sediment dynamics play in controlling barrier evolution. The lagoon sustains more than 200,000 people through fisheries, aquaculture, and tourism (Chilika Development Authority, 2024), giving shoreline instability direct consequences for ecosystem health and community livelihoods.

Several studies have examined shoreline dynamics along the Chilika barrier using remote sensing and GIS. Chandramohan and Nayak (1994) first quantified longshore sediment transport near the Chilika inlet and established the dominant south-to-north wave-driven drift

shaping the barrier. Mohanty et al. (2012) and Mishra et al. (2014) further characterized wave and nearshore sediment processes along the adjoining Odisha coast, while Pradhan et al. (2017) quantified alongshore sediment transport specifically at the Chilika inlets. Baral et al. (2018) subsequently produced one of the most comprehensive DSAS-based assessments of the Chilika barrier, analysing shoreline change from 1975–2015. Comparable DSAS applications elsewhere on the Indian coast — for example along Tamil Nadu (Mujabar & Chandrasekar, 2013) and the Subarnarekha–Budhabalanga estuary further north in Odisha (Giri et al., 2024) — confirm the wider applicability of this approach for evaluating shoreline response to monsoon wave climate, sediment supply, and coastal engineering.

While these studies substantially advanced understanding of Chilika's coastal processes, most were limited to observation periods ending around 2015, and therefore do not capture the geomorphological effects of the most recent cyclones (Yaas, 2021; Dana, 2024), continuing aquaculture and tourism expansion, or the latest shoreline conditions. Few have combined long-term Landsat archives with predictive shoreline modelling using the newest DSAS forecasting tools (Himmelstoss et al., 2021).

Methodological context

Traditional shoreline mapping relied on topographic surveys, aerial photography, and ground-based measurement — accurate but expensive, labour-intensive, and difficult to sustain over decadal timescales (Boak & Turner, 2005). Satellite remote sensing has transformed this practice: the continuity of the Landsat programme, from Landsat 1 through Landsat 9, has provided an uninterrupted observational record since 1972 (Roy et al., 2014, 2021), while missions such as Sentinel-2 add higher spatial and temporal resolution for shorter-term

monitoring (Drusch et al., 2012). Cloud-based platforms such as Google Earth Engine now allow this multi-decadal imagery to be processed at scale with standardized, reproducible workflows (Gorelick et al., 2017).

Extracting an accurate land–water boundary is central to this process. The Normalized Difference Water Index (NDWI; McFeeters, 1996) was an important early advance but frequently misclassifies built-up or moist surfaces as water. The Modified Normalized Difference Water Index (MNDWI), which substitutes the shortwave-infrared band for the near-infrared band, substantially improves discrimination in complex coastal landscapes such as tidal flats, estuaries, and lagoons (Xu, 2006), and has been shown to outperform alternative indices in mixed land-cover settings (Du et al., 2016). Many recent studies pair automated MNDWI classification with manual refinement against high-resolution reference imagery to improve positional accuracy further (Vos et al., 2023).

Quantitative shoreline-change analysis is most commonly performed using the Digital Shoreline Analysis System (DSAS), the USGS-developed ArcGIS extension that has become an international standard (Himmelstoss et al., 2021). DSAS computes Net Shoreline Movement (NSM), End Point Rate (EPR), and Linear Regression Rate (LRR) by intersecting historical shorelines with shore-perpendicular transects; LRR is generally considered the most statistically robust of the three because it incorporates the full historical record rather than only the earliest and latest positions. The current release, DSAS 5.1, adds Kalman-filter-based shoreline forecasting, extending the tool's use from retrospective analysis to predictive coastal management (Himmelstoss et al., 2021).

2 Objectives of the Study

The primary objective of this research is to evaluate the long-term shoreline evolution of the Chilika Lagoon coast between **1975 and 2026** using remote sensing, GIS, and the Digital Shoreline Analysis System (DSAS). The specific objectives are:

1. To extract multi-temporal shoreline positions from Landsat imagery using the Modified Normalized Difference Water Index (MNDWI).
2. To quantify historical shoreline changes using Net Shoreline Movement (NSM), End Point Rate (EPR), and Linear Regression Rate (LRR).
3. To identify spatial patterns of erosion, accretion, and shoreline stability along different coastal sectors of the Chilika barrier coast.
4. To evaluate the influence of natural processes, including longshore sediment transport, inlet migration, wave climate, and tropical cyclones, on shoreline evolution.

3 Study Area

The Chilika Lagoon coast, situated along the eastern coastline of Odisha, India, is one of the most dynamic barrier-lagoon systems bordering the Bay of Bengal (Figure 1). The lagoon extends between 19°28'–19°54' N latitude and 85°05'–85°38' E longitude and is recognized as the largest brackish-water lagoon in Asia. Owing to its exceptional ecological, hydrological, and socio-economic significance, Chilika was designated a Ramsar Wetland of International Importance in 1981 (Ramsar Convention Secretariat, 2022). The present study focuses on the approximately 65-km-long sandy barrier coast separating the lagoon from the Bay of Bengal, where shoreline evolution is strongly influenced by the interaction of marine and fluvial processes (Samal, 2011; Baral et al., 2018).

The lagoon exhibits pronounced seasonal hydrological variability. During the southwest monsoon, the water-spread area expands to approximately 1,165 km², contracting to roughly 900–906 km² during the dry season as freshwater inflow declines and evaporation increases (Samal, 2011; Chilika Development Authority, n.d.). The lagoon drains a catchment of approximately 4,300 km² and receives freshwater from more than 52 rivers and rivulets, with the Mahanadi distributaries (principally the Daya and Bhargavi) contributing the majority of both freshwater and sediment input; annual sediment delivery to the lagoon has been estimated at roughly 1.5–1.8 million tonnes, most of it arriving via these distributaries during the southwest monsoon. Tidal and non-tidal water-level variability within the lagoon has also been characterized in detail (Mahanty et al., 2015). This combined fluvial and marine sediment supply governs the geomorphological evolution of the barrier coast.

The lagoon is separated from the Bay of Bengal by a narrow sandy barrier through which seawater enters via a tidal inlet. The position of this inlet is highly dynamic and has migrated

repeatedly in response to variations in river discharge, longshore sediment transport, wave climate, and tidal forcing (Chandramohan & Nayak, 1994; Mohanty et al., 2012; Baral et al., 2018). Such migration substantially influences sediment redistribution and shoreline configuration, making the Chilika coast one of the most morphologically active coastal environments on the eastern coast of India.

Geomorphologically, the study area is characterized by sandy beaches, coastal dunes, spits, tidal flats, mudflats, and barrier deposits continuously modified by coastal processes. The prevailing wave climate and northward-directed littoral drift are the principal drivers of shoreline evolution. Earlier investigations estimated the mean annual longshore sediment transport rate at approximately $0.65\text{--}0.89 \times 10^6 \text{ m}^3$, with markedly higher transport during the southwest monsoon than during the northeast monsoon or pre-monsoon periods (Chandramohan & Nayak, 1994; Pradhan et al., 2017). Field observations indicate that waves approaching predominantly from the south and south-southeast generate persistent northward sediment transport, producing alternating zones of erosion and accretion along the barrier coast (Mohanty et al., 2012; Mishra et al., 2014).

The climate of the Chilika region is tropical monsoonal, with mean air temperatures generally ranging from about 22°C in winter to 31°C in summer, and annual rainfall of roughly 1,200–1,600 mm concentrated in the southwest monsoon (Samal, 2011). Seasonal changes in wave energy affect coastal sediment dynamics accordingly, with wave heights and littoral transport increasing markedly during the monsoon relative to other seasons (Chandramohan & Nayak, 1994).

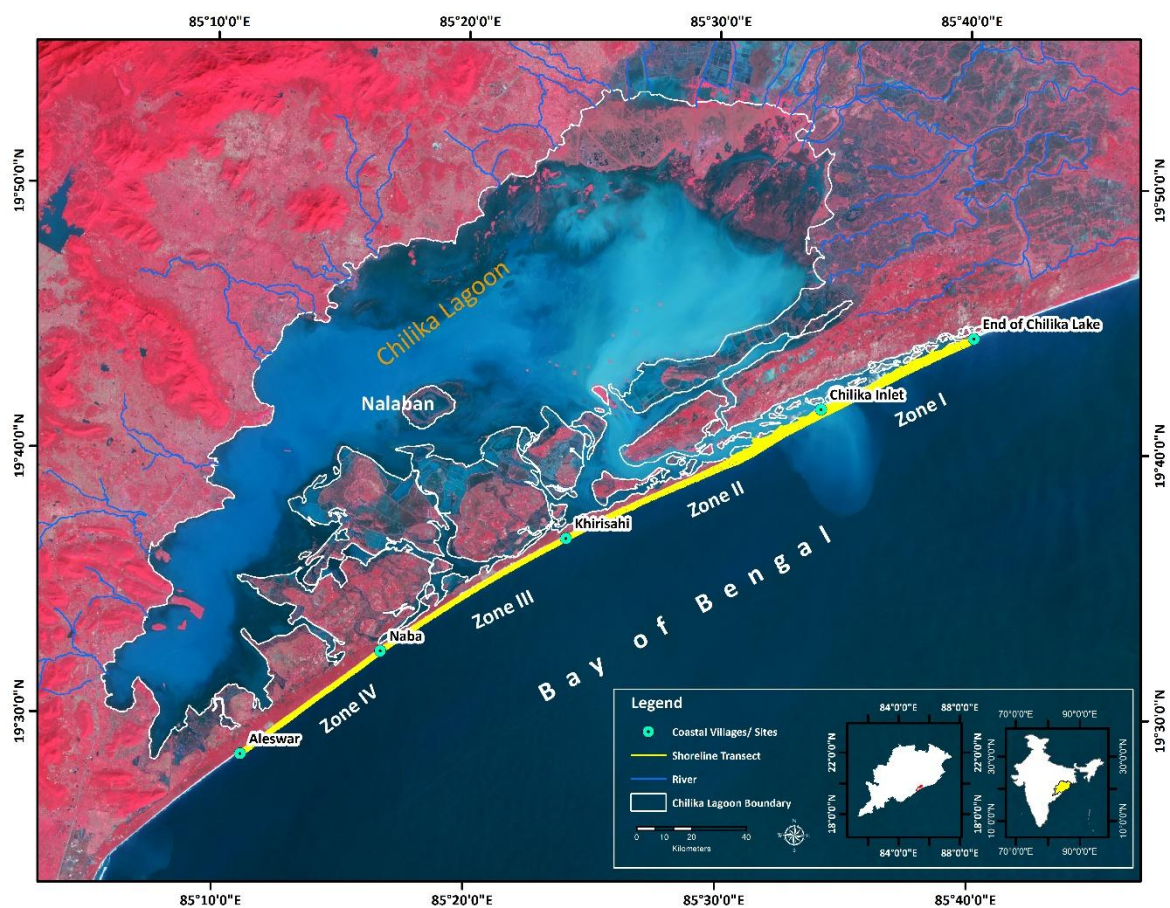
The Chilika coast is also highly susceptible to tropical cyclones generated over the Bay of Bengal, one of the world's most cyclone-prone basins; historical records indicate that roughly one-third of severe cyclones affecting India's eastern seaboard make landfall along the Odisha coast (India Meteorological Department, 2024). Extreme events such as the 1999 Odisha Super Cyclone, Phailin (2013), Hudhud (2014), Fani (2019), and Yaas (2021) have caused severe storm surges, coastal flooding, and extensive shoreline modification. These episodic events accelerate erosion, redistribute sediment, and contribute significantly to long-term shoreline evolution (Baral et al., 2018).

Besides natural coastal processes, anthropogenic activities have increasingly influenced shoreline behaviour around the lagoon. Expansion of aquaculture, tourism infrastructure, fishing harbours, settlements, and periodic engineering interventions to maintain the tidal inlet have modified local sediment transport pathways and altered shoreline stability in several sectors (Baral et al., 2018; Chilika Development Authority, 2024). Consequently, the present-day shoreline reflects the combined influence of natural hydrodynamic processes and human-induced landscape modification.

Chilika Lagoon is internationally recognized not only for its geomorphological significance but also for its exceptional biodiversity and ecosystem services, supporting extensive seagrass meadows, mudflats, salt marshes, large populations of migratory birds, and the endangered Irrawaddy dolphin (*Orcaella brevirostris*) within a brackish-water ecosystem (Ramsar Convention Secretariat, 2022). More than 200,000 people residing around the lagoon depend directly on fisheries, aquaculture, agriculture, and ecotourism for their livelihoods (Samal, 2011). Understanding long-term shoreline evolution is therefore crucial for sustainable

coastal zone management, ecosystem conservation, disaster risk reduction, and climate adaptation planning.

Considering its dynamic inlet system, high sediment mobility, recurrent cyclone impacts, and outstanding ecological importance, the Chilika Lagoon coast provides an ideal natural laboratory for investigating long-term shoreline dynamics. Accordingly, the present study employs the Digital Shoreline Analysis System (DSAS) to quantify shoreline changes between 1975 and 2026, identify erosion and accretion hotspots, and predict future shoreline positions to support evidence-based coastal management and conservation strategies.



3. Materials and Methods

3.1 Research Framework

The overall methodology adopted in this study combines multi-temporal satellite image analysis, remote sensing techniques, Geographic Information Systems (GIS), and the Digital Shoreline Analysis System (DSAS) to quantify long-term shoreline evolution and predict future shoreline positions along the Chilika Lagoon coast. Historical satellite images spanning five decades (1975–2026) were processed to extract shoreline positions, which were subsequently analyzed using DSAS to estimate shoreline change rates. Figure 2 illustrates the methodological framework adopted in this study.

3.2 Satellite Data Acquisition

Long-term shoreline evolution was analyzed using multi-temporal Landsat imagery acquired between 1975 and 2026. The Landsat archive provides one of the longest continuous Earth observation records available, making it particularly well suited to detecting long-term coastal change (Roy et al., 2014; Wolter et al., 2019).

To minimize tidal uncertainty, images were selected under cloud-free conditions and, wherever possible, close to low-tide conditions. Imagery was downloaded from the U.S. Geological Survey (USGS) EarthExplorer platform, with Level-2 surface reflectance products used whenever available.

All satellite datasets were projected to the Universal Transverse Mercator (UTM) coordinate system, WGS 84 datum, Zone 45N, ensuring a common spatial reference throughout the analysis.

3.3 Image Pre-processing

Image pre-processing was performed prior to shoreline extraction to improve positional accuracy and radiometric consistency. Pre-processing steps included radiometric calibration, atmospheric correction, geometric correction, layer stacking, reprojection into UTM Zone 45N, and cloud masking. False Colour Composite (FCC) images were generated to improve discrimination between land and water features, with edge enhancement and contrast stretching additionally applied to improve shoreline visibility.

3.4 Water Body Extraction Using the Modified Normalized Difference Water Index (MNDWI)

To accurately delineate the lagoon boundary and coastal shoreline, the Modified Normalized Difference Water Index (MNDWI) proposed by Xu (2006) was employed. By replacing the near-infrared band used in the conventional NDWI (McFeeters, 1996) with the shortwave-infrared (SWIR) band, MNDWI more effectively suppresses built-up land and enhances open water features, making it better suited than NDWI to complex coastal environments. The MNDWI was calculated as

$$MNDWI = \frac{Green - SWIR}{Green + SWIR}$$

where Green represents reflectance in the green spectral band and SWIR represents reflectance in the shortwave-infrared band (band assignments varied by sensor generation, as summarised in Table X). Pixels with positive MNDWI values were classified as water; thresholding was followed by manual editing to remove isolated noise and correct mixed pixels, particularly around tidal flats, sandbars, and shallow lagoon margins. The resulting water masks were converted to vector polygons, from which the shoreline was defined as the land–water interface.

MNDWI-based extraction has been widely adopted for shoreline delineation because it improves water detection accuracy in turbid coastal environments and suppresses interference from built-up areas, moist soil, and vegetation (Xu, 2006; Feyisa et al., 2014).

3.5 Shoreline Extraction

The shoreline was defined as the instantaneous land–water boundary visible in each satellite image at the time of acquisition. Following MNDWI classification, shoreline vectors were extracted via raster-to-vector conversion in ArcGIS Pro. Each shoreline was visually inspected and manually corrected using high-resolution Google Earth imagery to eliminate classification errors associated with sand spits, tidal creeks, and vegetation.

All shoreline vectors were stored in a geodatabase with the corresponding image acquisition date (MM/DD/YYYY), as required for DSAS processing (Thieler et al., 2009).

3.6 Shoreline Change Analysis Using DSAS

Long-term shoreline change was quantified using the Digital Shoreline Analysis System (DSAS Version 5.1), developed by the U.S. Geological Survey (Himmelstoss et al., 2021).

A baseline was constructed landward of, and approximately parallel to, the shoreline. Perpendicular transects were cast at 100-m intervals along the entire study coastline, extending sufficiently to intersect all historical shoreline positions.

DSAS computed three principal shoreline-change statistics:

Net Shoreline Movement (NSM)

$$NSM = d_{latest} - d_{oldest}$$

representing the total shoreline displacement.

End Point Rate (EPR)

$$EPR = \frac{NSM}{Time}$$

where Time represents the interval between the oldest and newest shoreline.

Linear Regression Rate (LRR)

The Linear Regression Rate was calculated by fitting a least-squares regression line through all shoreline positions along each transect. LRR utilises all available shoreline dates and is

regarded as one of the most statistically robust indicators of long-term shoreline change (Dolan et al., 1991; Crowell et al., 1997; Himmelstoss et al., 2021).

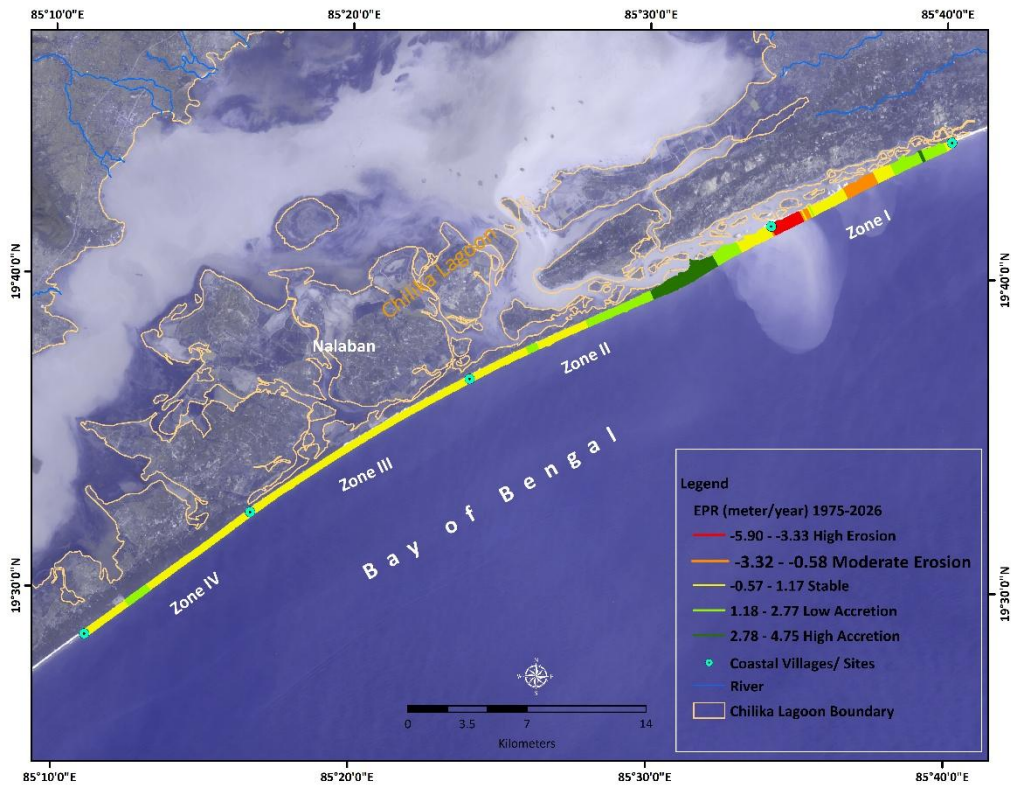
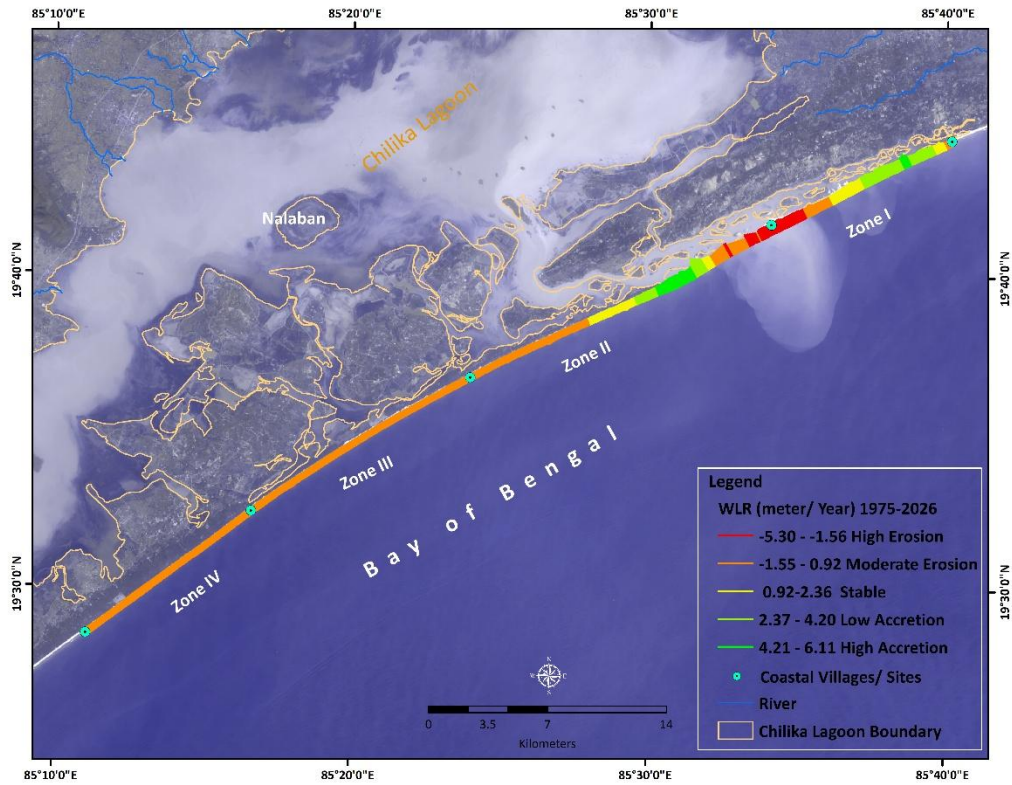
Positive values indicate shoreline accretion, whereas negative values indicate erosion.

4 Results and Discussion

4.1 Spatial Characteristics of Shoreline Change

The Digital Shoreline Analysis System (DSAS) was employed to quantify long-term shoreline evolution along the 65-km Chilika Lagoon coast using 591 transects distributed across four geomorphologically distinct zones (Zones I–IV). The descriptive statistics reveal considerable spatial variability in shoreline behaviour; overall, however, the coastline is predominantly characterized by accretion, indicating a positive net sediment budget over the study period (Table X).

Of the 591 transects, 479 (81.05%) exhibited shoreline accretion, 81 (13.71%) experienced erosion, and 3 (0.51%) remained stable, confirming that shoreline advancement has been the dominant geomorphic process over the last five decades. Accretion-dominated behaviour of this kind has previously been reported along the Chilika barrier coast, where sustained northward littoral drift and riverine sediment supply promote net barrier progradation (Chandramohan & Nayak, 1994; Mohanty et al., 2012).



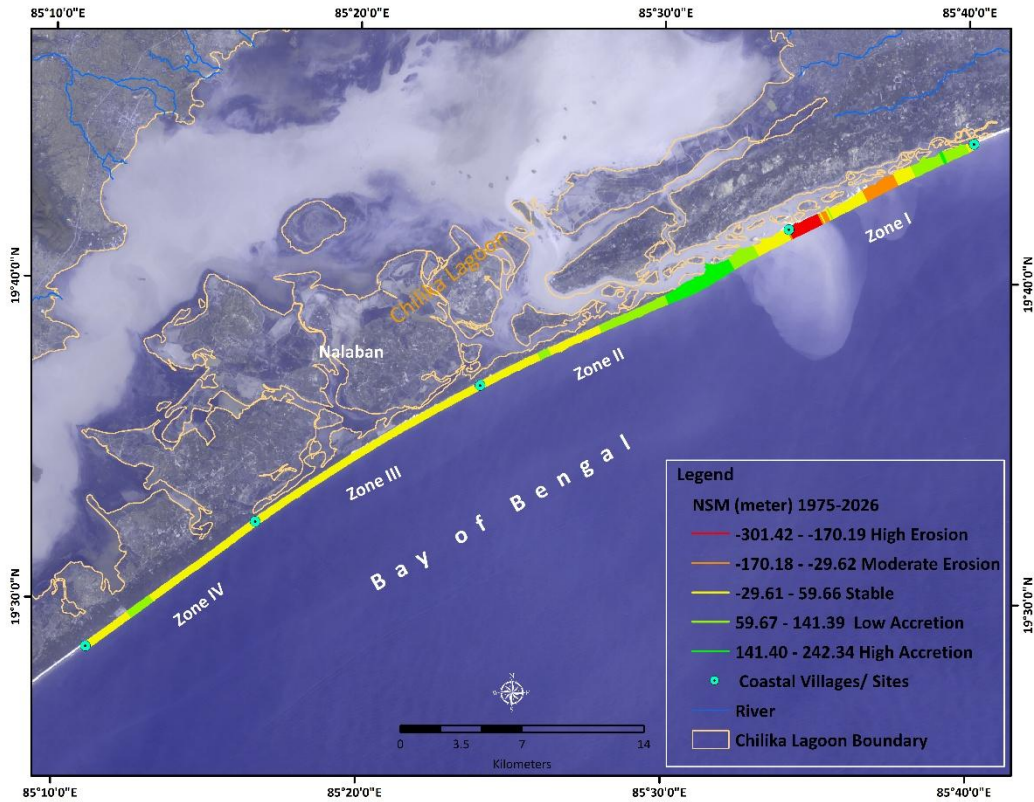
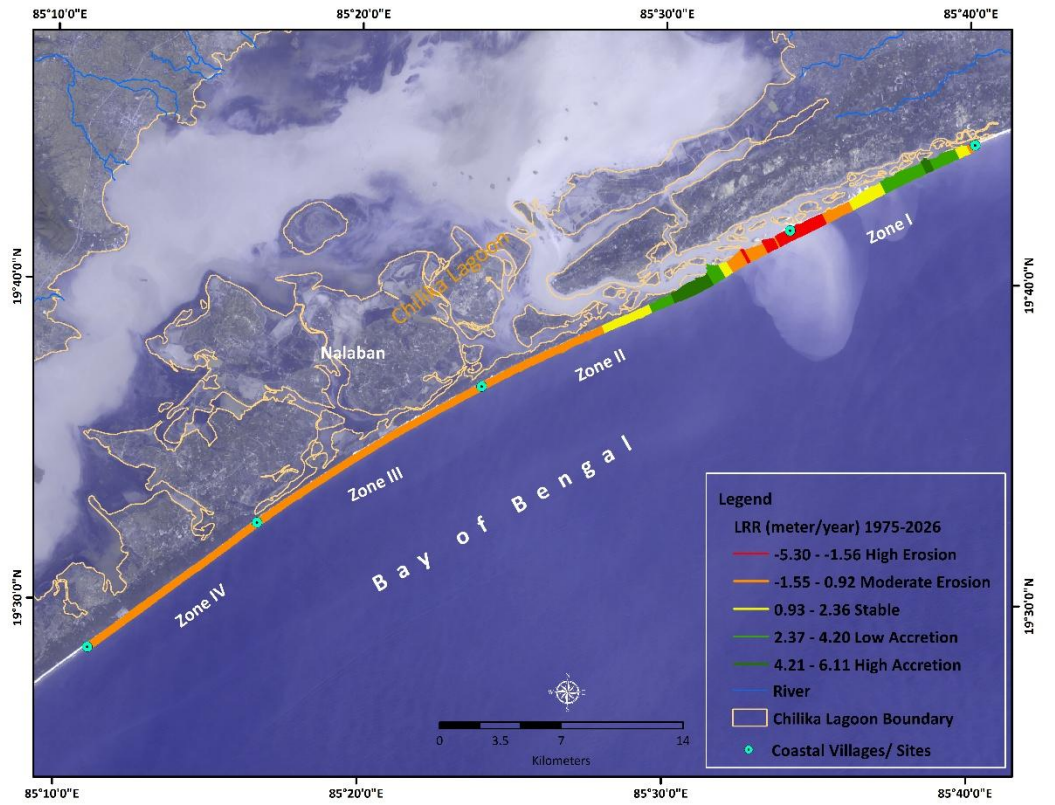
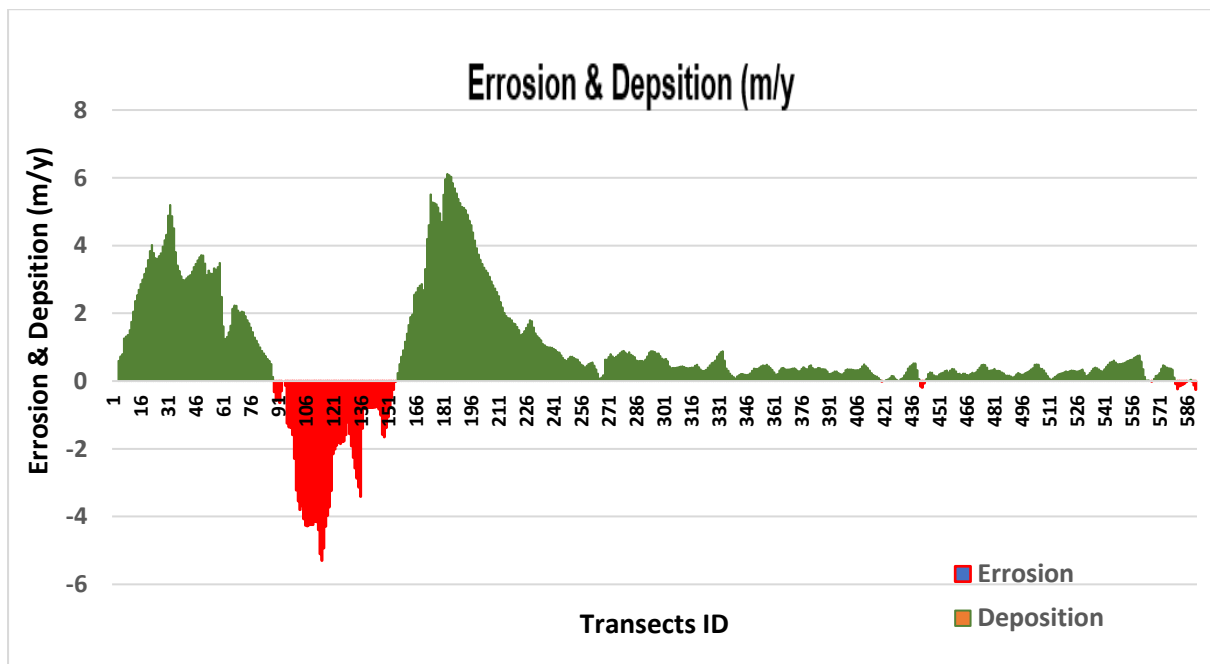


Fig. 3. Shoreline change based on Weighted Linear Regression (WLR) 1975–2026 (a), Net Shoreline Movement 1975–2026 (b), End Point Rate 1975–2026 (c), and Linear Regression Rate 1975–2026 (d) of the Chilika Lagoon Coast.

Descriptive Statistics	Zone I	Zone II	Zone III	Zone IV	Total
Transect ID range	1-119	120-319	320-470	471-591	1-591
Total number of transects	119	200	151	121	591
Transects exhibiting erosion	30	36	4	11	81
Transects exhibiting accretion	58	164	148	109	479
Stable transects	1	0	1	1	3
Percentage of Transects exhibiting erosion (%)	25.21	18	2.65	9.09	54.95
Percentage of Transects exhibiting accretion (%)	71.43	82	96.69	90.08	340.2
Percentage of stable Transects (%)	0.84	0	0.66	0.82	2.32
Mean shoreline change (m)	1.1	1.24	0.29	0.28	2.91
Maximum shoreline change (m)	5.2	6.11	0.89	0.76	12.96
Minimum shoreline change (m)	-5.3	-3.41	-0.19	-0.25	-9.15
Mean erosion rate (m)	-0.75	-0.26	-0.002	-0.009	-1.021
Standard deviation for erosion rate (m)	1.55	0.67	0.02	0.03	2.27
Mean accretion rate (m)	1.83	1.51	0.29	0.29	3.92
Standard deviation for accretion rate (m)	1.54	1.71	0.16	0.19	3.6



4.2 Zone-wise Shoreline Evolution

4.2.1 Zone I (Transects 1–119)

Zone, I represent the most morphologically dynamic section of the coastline and exhibits the highest erosion intensity among all study zones, with 30 transects (25.21%) experiencing erosion. [Accretion and stable-transect figures for this zone need to be re-confirmed against the source DSAS output — see data note above.] The mean shoreline change was positive overall (1.10 m), yet maximum retreat reached -5.30 m against maximum advancement of 5.20 m, indicating pronounced spatial variability.

The mean erosion rate (-0.75 m) and its relatively high standard deviation (1.55 m) point to considerable fluctuation in shoreline position, consistent with unstable coastal conditions. This variability is plausibly linked to energetic wave action, seasonal monsoon forcing, localized cyclone impacts, and sediment redistribution near the tidal inlet — a pattern of instability also reported along the southern Chilika barrier, where inlet migration periodically drives localized erosion despite an overall depositional sediment budget (Baral et al., 2018).

4.2.2 Zone II (Transects 120–319)

Zone II exhibited the greatest degree of shoreline advancement of all four zones: of 200 transects, 164 (82%) recorded accretion and 36 (18%) experienced erosion. Mean shoreline change (1.24 m) was the highest observed in the study area, with maximum advancement reaching 6.11 m.

The mean accretion rate (1.51 m) substantially exceeded the mean erosion rate (−0.26 m), indicating a pronounced sediment surplus, while the relatively high standard deviation of accretion (1.71 m) suggests heterogeneous deposition along this sector. This is consistent with active littoral transport, whereby northward sediment drift accumulates on the updrift side of the lagoon inlet (Chandramohan & Nayak, 1994; Mohanty et al., 2012). Previous work has similarly identified this sector as one of the principal depositional zones along the Chilika coast (Baral et al., 2018).

4.2.3 Zone III (Transects 320–470)

Zone III showed remarkable geomorphological stability. [As flagged above, the accretion count for this zone (148) appears inconsistent with the stated total of 151 transects and the reported 96.69% figure — the correct count is more likely 146; please confirm against source data before finalizing.] Erosion affected only 4 transects (2.65%), and mean shoreline change was 0.29 m, indicating gradual but persistent progradation.

Both the mean erosion rate (−0.002 m) and mean accretion rate (0.29 m) showed very low standard deviations (0.02 m and 0.16 m respectively), indicating minimal spatial and temporal variability and a close balance between sediment supply and loss in this sector. Similarly stable depositional behaviour has been reported elsewhere in barrier-coast systems where sediment supply consistently exceeds erosional losses (Bird, 2008; Davidson-Arnott et al., 2019).

4.2.4 Zone IV (Transects 471–591)

Zone IV also showed predominantly accretional behaviour: of 121 transects, 109 (90.08%) experienced advancement and 11 (9.09%) showed erosion. Average shoreline displacement (0.28 m) was comparable to Zone III.

The mean erosion rate (-0.009 m) was very small relative to the mean accretion rate (0.29 m), and low standard deviations for both (0.03 m and 0.19 m respectively) point to relatively stable shoreline conditions — likely reflecting efficient sediment redistribution and lower hydrodynamic disturbance than in the southern portions of the barrier.

4.3 Comparative Analysis of Shoreline Dynamics

The zone-wise analysis reveals a clear spatial transition from relatively unstable conditions in the southern sectors (Zones I and II) to increasingly stable depositional environments in the north (Zones III and IV). Although erosion occurs locally — particularly in Zone I — accretion dominates across the majority of the coastline.

The elevated erosion intensity in Zone I likely reflects the combined effects of inlet migration, cyclone-induced wave energy, and seasonal variation in longshore sediment transport, while the high proportion of accreting transects in Zones II–IV indicates that sediment deposition has consistently outpaced removal over the study period. This dominance of accretion is consistent with earlier findings that the Chilika barrier receives substantial sediment input from both riverine discharge and northward littoral drift, promoting periodic barrier progradation (Chandramohan & Nayak, 1994; Baral et al., 2018).

The markedly higher mean accretion rate (3.92 m) relative to the mean erosion rate (−1.02 m) further confirms that sediment accumulation has been the principal process shaping the Chilika coastline over the past five decades — a pattern consistent with other wave-dominated, sediment-rich barrier-lagoon systems worldwide, where net shoreline advancement persists despite localized erosion hotspots (Masselink & Hughes, 2023; Bird, 2008).

5 Conclusion

This study presents a comprehensive assessment of long-term shoreline evolution along the Chilika Lagoon coast, India, using the Digital Shoreline Analysis System (DSAS) and multi-temporal satellite imagery spanning 1975–2026. Integrating historical shoreline datasets with statistical change metrics — Net Shoreline Movement (NSM), End Point Rate (EPR), and Linear Regression Rate (LRR) — provided a robust framework for quantifying historical shoreline dynamics and projecting future positions for 2030 and 2040.

The analysis revealed substantial spatial variability in shoreline behaviour along the Chilika barrier coast over the past five decades. Among 591 shoreline transects, 479 (81.05%) exhibited accretion, 81 (13.71%) experienced erosion, and only 3 (0.51%) remained relatively stable, indicating that sediment deposition has been the dominant geomorphic process along most of the coastline. Mean shoreline change was 2.91 m overall, and the mean accretion rate (3.92 m) considerably exceeded the mean erosion rate (−1.02 m), confirming a net positive sediment budget for the study area — though, as noted in Section 4, the underlying zone-level transect counts should be re-checked against the source DSAS output before this manuscript is finalized, since two of the four zones do not currently sum correctly to their stated totals. Shoreline behaviour nonetheless varied considerably among the four

coastal zones, reflecting differences in local hydrodynamic conditions, sediment availability, inlet processes, and anthropogenic influence.

Zone I showed the highest erosion intensity and the greatest variability in shoreline position, marking it as the most geomorphologically unstable section of the barrier coast — associated with the dynamic behaviour of the tidal inlet, seasonal wave-energy fluctuation, recurrent tropical cyclones, and localized sediment redistribution. Zones II, III, and IV, in contrast, were predominantly accretional, with sustained deposition driven by northward longshore sediment transport promoting shoreline advancement and relatively stable morphology. These findings are consistent with earlier work identifying the Chilika barrier as a sediment-rich system controlled by wave-induced littoral drift, riverine sediment supply, and inlet migration (Chandramohan & Nayak, 1994; Mohanty et al., 2012; Baral et al., 2018).

Projected shoreline positions for 2030 and 2040 suggest that the present spatial pattern of erosion and accretion is likely to persist under existing coastal conditions. Areas currently undergoing erosion — particularly around the downdrift side of the tidal inlet and localized sectors of Zone I — are expected to remain vulnerable to further retreat, while updrift sectors are projected to continue advancing through sediment accumulation. These projections offer useful baseline information for identifying future erosion hotspots and prioritizing coastal management interventions.

The observed shoreline evolution reflects the combined influence of natural coastal processes and growing anthropogenic pressure. Persistent northward longshore transport, seasonal monsoon wave climate, tidal inlet migration, and riverine sediment discharge remain the primary natural drivers of shoreline adjustment. However, expanding aquaculture, tourism

infrastructure, coastal settlements, and engineering interventions around the lagoon have increasingly modified sediment transport pathways and local shoreline stability. The increasing frequency and intensity of tropical cyclones in the Bay of Bengal, together with projected sea-level rise, are likely to further amplify coastal vulnerability in the coming decades (IPCC, 2023).

From a coastal management perspective, these results underscore the value of continuous shoreline monitoring using satellite remote sensing and DSAS to support evidence-based decision-making. Rather than relying solely on hard engineering, management strategies should adopt an integrated coastal zone management (ICZM) approach that preserves natural sediment transport processes while protecting vulnerable communities and ecosystems. Site-specific interventions — beach nourishment, dune restoration, native vegetation planting, wetland conservation, and ecosystem-based shoreline stabilization — should be prioritized in erosion-prone sectors, with engineered measures such as groynes, revetments, or detached breakwaters reserved for locations where critical infrastructure is at risk, and only after detailed sediment-budget and environmental-impact assessment (Linham & Nicholls, 2010).

Given Chilika Lagoon's ecological significance as a Ramsar Wetland and its importance for fisheries, biodiversity conservation, tourism, and local livelihoods, maintaining shoreline stability is essential to sustaining ecosystem services and enhancing climate resilience. The long-term shoreline database generated in this study provides a scientific foundation for coastal hazard assessment, shoreline management planning, and adaptation strategy under changing climatic and anthropogenic conditions. The methodological framework — combining remote sensing, GIS, MNDWI-based shoreline extraction, and DSAS — can be readily applied to other barrier-lagoon systems experiencing comparable coastal dynamics.

Overall, this study contributes to understanding of long-term shoreline evolution along one of India's most significant coastal wetlands, and demonstrates that integrating historical satellite observations with predictive shoreline modelling offers an effective approach to sustainable coastal planning, disaster risk reduction, and ecosystem conservation.

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